The Hexagonality Law and the Fibonacci Stacking

The golden ratio in the volumes of a never-repeating crystal

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. DICTO's long-standing aim is a bridge between repeating and never-repeating order. Close-packed layers can be stacked in a never-repeating Fibonacci order: layer types cubic (c) and hexagonal (h) following the substitution c → ch, h → c. We prove that in this stacking the share of hexagonal layers is exactly $1/\varphi^2$ and that hexagonal layers never touch, so of the four kinds of piece between neighbouring layers only three occur, in the golden proportions $1/\varphi^3 : 1/\varphi^2 : 1/\varphi^2$ (cubic–cubic, cubic–hexagonal, hexagonal–cubic). By the Hexagonality Law (#66), every kind of piece then averages exactly $V(cc) + (V(hh) - V(cc))/\varphi^2$ per layer: the golden ratio enters the volumes of a crystal exactly. In rock salt at weight $\lambda = 1/3$ the ball piece averages $10999/20736 - (71/6912)\sqrt{5}$ per layer. The Fibonacci order closes into periodic stackings of 15, 16, 26 and 63 layers with hexagonalities 2/5, 3/8, 5/13 and 8/21, ratios of Fibonacci numbers. On the first three, the Laves phases keep the law exactly in all 90 tests, including the foot atom for which the law is not proved. The stackings in which hexagonal layers never touch are exactly the configurations of the supersymmetric lattice fermions of Fendley, Schoutens and de Boer, whose ground states have the hexagonality of 6H.

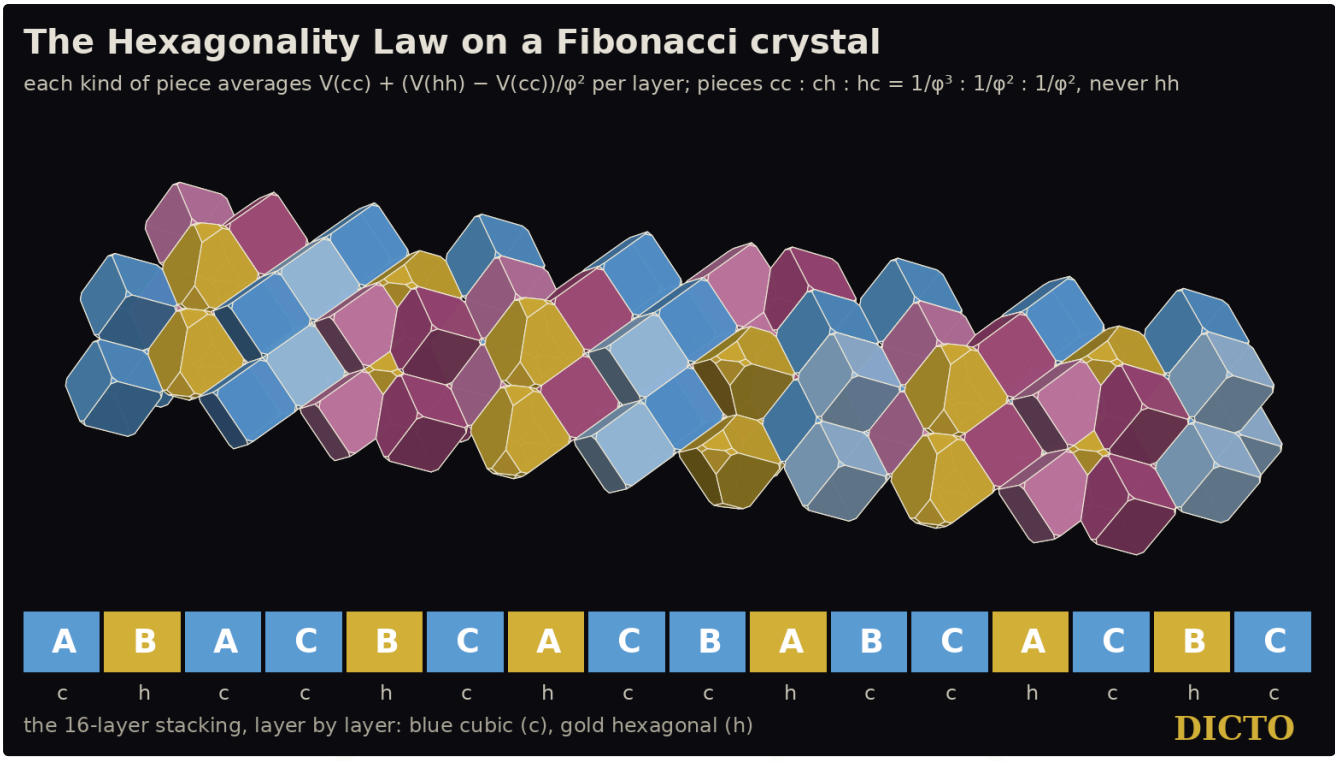


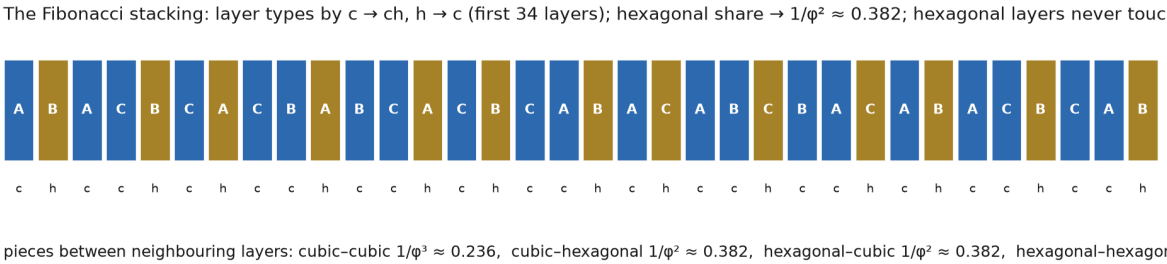
Figure 1. The Hexagonality Law on a Fibonacci crystal: rock salt on the 16-layer Fibonacci stacking ABACBCACBABCACBC at weight λ = 1/3, three neighbouring columns, the stacking running left to right. Each piece is the exact power cell of its site. Balls: cubic layers blue, hexagonal layers gold. Octahedral atoms between layers: cubic–cubic pale blue, cubic–hexagonal and hexagonal–cubic rose; hexagonal–hexagonal never occurs. Volumes agree with the exact run to nine decimals.

1. From repeating to never-repeating

A stacking of close-packed layers is a sequence of positions A, B, C, each layer in one of the two positions not taken by the layer below. The pure stackings repeat: ABC (cubic) and AB (hexagonal). The rule of halves (#57), the Swap Rule (#61, #64) and the Hexagonality Law (#66) hold on every stacking, and #67 drew from them that the totals of a random stacking are fixed by its share of hexagonal layers alone. DICTO's aim, to bridge the periodic and the aperiodic, asks for the most ordered of the never-repeating stackings: the Fibonacci one. In *A Fibonacci Spine* (#30) the Fibonacci chain appeared in a five-fold arrangement of the DICTO Ogdoecontahedron; here it is the order of the layers themselves.

2. The Fibonacci stacking

A layer is cubic (c) when its two neighbours sit in different positions and hexagonal (h) when they sit in the same (Jagodzinski). Any sequence of types is a stacking: with steps $s_k = \pm 1$ between layers, layer k is c when $s_{k-1} = s_k$ and h when they differ, so the types fix the steps up to one sign. The **Fibonacci stacking** takes the types in the order of the Fibonacci word, the limit of the substitution c → ch, h → c started from c: chcchchcchcch… (Figure 2).



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Figure 2. The first 34 layers of the Fibonacci stacking, coloured by type (blue cubic, gold hexagonal), with the position A, B, C of each layer. The hexagonal layers never touch.

> **Theorem.** In the Fibonacci stacking (i) the share of hexagonal layers is $h = 1/\varphi^2 = (3 - \sqrt{5})/2$; (ii) no two hexagonal layers are neighbours; (iii) among the pairs of neighbouring layers, cubic–cubic has frequency $1/\varphi^3$, cubic–hexagonal and hexagonal–cubic $1/\varphi^2$ each, hexagonal–hexagonal 0. (iv) For every kind of piece to which the Hexagonality Law applies, the average volume per layer is exactly $V(cc) + (V(hh) - V(cc))(3 - \sqrt{5})/2$.

Proof. (i) The substitution has matrix [[1, 1], [1, 0]] on the counts (c, h); its eigenvalue φ gives letter frequencies $1/\varphi$ for c and $1/\varphi^2$ for h, the classical result for the Fibonacci word. (ii) Every h in a substituted word is the image of c → ch, the second letter of a block, and every block begins with c; so each h is followed by c, and hh never occurs. (iii) Each h is therefore preceded and followed by c, so ch and hc each have the frequency of h, $1/\varphi^2$, and cc has $1 - 2/\varphi^2 = 1/\varphi^3$ (using $\varphi^2 = \varphi + 1$). (iv) By #66, over any long run of layers the total of such a piece is $n_c V(cc) + n_h V(hh)$ up to two end terms (#67, Theorem 1); divide by the number of layers and use (i). ∎

So a Fibonacci crystal contains the pure cubic piece and the two mixed pieces only, in golden proportions, and never the pure hexagonal piece; and the pieces of each kind average to a number in $\mathbb{Q}(\sqrt{5})$.

3. Rock salt in the Fibonacci order

In rock salt and nickel arsenide (balls and octahedral atoms of weight λ, tetrahedral holes empty, #64), at $\lambda = 1/3$ the exact averages per layer are:

Piece	Average per layer, exact	Decimal
ball	$10999/20736 - (71/6912)\sqrt{5}$	0.507461
octahedral atom	$9715/20736 + (215/20736)\sqrt{5}$	0.491693
each tetrahedral hole	$11/20736 - (1/20736)\sqrt{5}$	0.000423

They add up to the volume of one layer, as they must; the √5 parts cancel in the sum.

4. Repeating approximants, computed exactly

A periodic stacking must close: an even number of sign changes, and steps adding up to a multiple of 3. The Fibonacci words of Fibonacci lengths close after repeating them once, twice or three times: 5 types (chcch) three times gives the 15-layer stacking ABACBCACBABCBAC; 8 types twice the 16-layer ABACBCACBABCACBC; 13 types twice the 26-layer ABACBCACBABCACBCABACABCBAC; 21 types three times a 63-layer stacking. Their hexagonalities 2/5, 3/8, 5/13, 8/21 are ratios of Fibonacci numbers and tend to $1/\varphi^2$.

For the pieces fixed by their two neighbouring layers the law is a theorem (#66), so these stackings follow it. The Laves foot atom M0 reads three layers above $\lambda = 17/96$, where the law is not proved, and #67 found it exact on every stacking up to eight layers. We walked the Laves phases exactly on the 15-, 16- and 26-layer Fibonacci approximants with #57's census engine (all census checks passed) and tested every role in every window of λ against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$.

> **Result.** On the 15-, 16- and 26-layer Fibonacci approximants, every role of the Laves phases, the foot atom M0 included, keeps the Hexagonality Law exactly: 90 of 90 tests. The 26-layer approximant (hexagonality $5/13 \approx 0.385$) is the closest to the golden $1/\varphi^2$ computed so far.

5. Hard-core fermions

Read a hexagonal layer as a particle on a ring of sites, one site per layer. A stacking in which no two hexagonal layers touch, the Fibonacci stacking among them, is then a configuration of particles with hard cores: never two on neighbouring sites. These are exactly the configurations of the supersymmetric lattice fermions of Fendley, Schoutens and de Boer (2003), whose Hamiltonian is $H = \{Q^+, Q^-\}$ with fermions that may not sit next to each other. On the periodic chain of N sites their ground states have $\mathrm{int}((N + 1)/3)$ fermions, a filling of 1/3, two of them when N is a multiple of 3. We counted, for every N from 2 to 15, the hard-core configurations by particle number: they agree with theirs (for N = 6: 1, 6, 9 and 2 with 0 to 3 particles, total 18), and so does their Witten index, $2(-1)^p$ for $N = 3p$ and $(-1)^p$ otherwise.

A crystal adds two conditions, because a periodic stacking must close: the step signs close only for an even number of hexagonal layers, so stackings lie in the sector of even particle number, and the layer positions close only when the steps add up to a multiple of 3. For N = 6 the closing configurations are the cubic stacking (no particle) and the three positions of 6H (two particles, h c c h c c); and 6H, the commonest polytype of silicon carbide, has hexagonality 1/3, the ground-state filling of the supersymmetric chain, in the same particle-number sector. The two pictures differ in their physics: the supersymmetric fermions hop and repel, while by the Hexagonality Law the volume each hexagonal layer carries does not depend on the others. We record the correspondence of configurations and counts only; the Fibonacci stacking's density $1/\varphi^2 \approx 0.382$ lies between the ground-state filling 1/3 and the hard-core limit 1/2.

6. Conclusion

The Fibonacci stacking is the never-repeating crystal of close-packed layers in its most ordered form. Its share of hexagonal layers is $1/\varphi^2$, its hexagonal layers never touch, and its pieces come in the golden proportions $1/\varphi^3 : 1/\varphi^2 : 1/\varphi^2$. By the Hexagonality Law the golden ratio then enters every average piece volume exactly. One kit of pieces, borrowed from the two repeating stackings, builds the never-repeating one: the bridge between the two orders that DICTO set out to find, here at the level of the pieces themselves.

7. Prior art and credit

The Fibonacci chain is the most studied never-repeating one-dimensional order: Baake (1999) obtains it by cutting and projecting the square lattice along a line of slope $1/\tau$, and calls it the

most common and best studied non-periodic 1D point set; its theory is set out in Baake and Grimm, *Aperiodic Order* (2013), and its Fibonacci numbers and words in the combinatorics of Lothaire (as cited by Kramer, 2011). Our sequence of layer types is this chain. Periodic structures converging to a quasicrystal are its approximants (Goldman and Kelton, 1993), which our 15-, 16-, 26- and 63-layer stackings are; a fault in a Fibonacci stacking is the local flip by which an approximant and its quasicrystal differ (Kléman, 2013). Hard-core lattice fermions with N = 2 supersymmetry are Fendley, Schoutens and de Boer (2003). Disordered stackings of layers are described by Markov chains (Hart, Hansen and Kuhs, 2021; Ganesh and Nasr, 2023). The Fibonacci word and its letter and pair frequencies are classical. Fibonacci chains have been used to describe faulted close-packed crystals: Farkas-Jahnke proposed approximating heavily faulted structures such as ZnS by Fibonacci chains of five-layer structure elements (IUCr congress abstract). Fibonacci multilayers (for example quasiperiodic Fe/Cr magnetic multilayers) are studied in physics. We found no statement that a Fibonacci order of the layer types gives the golden proportions of the stacking's pieces, or exact golden-ratio volumes. The Fibonacci chain in DICTO's own work is #30; the laws used are #57, #61, #64, #66 and #67.

> **Version 2** (28 October 2026): the 26-layer approximant added (the law exact in all 90 tests); new §5 on hard-core supersymmetric fermions.

> **Sources checked** (28 October 2026). For Version 2: P. Fendley, K. Schoutens and J. de Boer, Lattice models with N = 2 supersymmetry, Phys. Rev. Lett. 90, 120402 (2003), arXiv hep-th/0210161, read (added to the reference library). Reference library, second pass (ask-library: "Fibonacci chain one-dimensional quasicrystal diffraction"; "approximants of quasicrystals Fibonacci periodic approximant phason"; "stacking disorder Markov chain layers hexagonality cubicity"; "Sturmian sequence golden mean substitution frequencies"; "aperiodic order Baake Grimm"): M. Baake, A guide to mathematical quasicrystals (1999), pp. 8–10 read (cut and project, the Fibonacci chain); P. Kramer, Gateways towards quasicrystals (2011), p. 7 (Fibonacci numbers, Lothaire); A. I. Goldman and K. F. Kelton, Quasicrystals and crystalline approximants (1993) and M. Baake and U. Grimm, Aperiodic Order, vol. 1 (2013), catalogue entries; M. Kléman, Defects in quasicrystals, revisited (2013), first page; A. G. Hart, T. C. Hansen and W. F. Kuhs (2021) and R. Ganesh and A. K. Nasr (2023), catalogue abstracts; S. Akiyama et al., Aperiodic tile sets from Sturmian lattices (2026) and M. Baake, D. Damanik and U. Grimm, What is aperiodic order? (2015), catalogue entries. First pass (ask-library "Fibonacci stacking sequence quasiperiodic close-packed layers polytype"): S. Dong, Fibonacci and Lucas sequences in aperiodic monotile supertiles (2024); P. J. Steinhardt and H.-C. Jeong (1996); D. Kamenetsky (2026); C. S. Kaplan (2025); M. Kléman (2013), catalogue entries and pages, none on stackings. Web (search summaries): M. Farkas-Jahnke, IUCr congress abstract E0552 (Fibonacci chains for faulted close-packed lattices); arXiv 1204.5523 and 2307.16052 (Fibonacci magnetic multilayers); arXiv 1012.4880; Int. Tables Vol. C ch. 9.2 (close packing).

> **Data and code.** Study 2026-10-28-fibonacci-stackings (fib_words.py, golden.py, golden_rocksalt.json, hexlaw_fib.py, fig69.py), using the exact weight walks of study 2026-10-27-splash-rules (run2; the new walks lvw_ABACBCACBABCBAC, lvw_ABACBCACBABCACBC, lvw_ABACBCACBABCACBCABACABCBAC) and study 2026-10-28-susy-fermions (count.py, counts.json), attached to this record. References: DICTO, A Fibonacci Spine (#30), DICTO DOI 10.5281/zenodo.23396361; The Focus (#66), 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636.